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Roll no: 30

CODE

```
import pandas as pd

# Create a DataFrame
data = {
    "Name": ["Amit", "Sneha", "Rahul", "Priya"],
    "Age": [20, 21, 19, 22],
    "Marks": [85, 90, 75, 88]
}

df = pd.DataFrame(data)

print("Original Data:")
print(df)

# 1. Display a column
print("\nName Column:")
print(df["Name"])

# 2. Add a new column
df["Grade"] = ["A", "A+", "B", "A"]

print("\nAfter Adding Grade Column:")
```

```
print(df)

# 3. Add a new row
df.loc[len(df)] = ["Neha", 20, 92, "A+"]

print("\nAfter Adding New Row:")
print(df)

# 4. Update data
df.loc[0, "Marks"] = 89

print("\nAfter Updating Marks:")
print(df)

# 5. Delete a column
df = df.drop("Grade", axis=1)

print("\nAfter Deleting Grade Column:")
print(df)

# 6. Filter data
print("\nStudents with Marks greater than 85:")
print(df[df["Marks"] > 85])

# 7. Sort data
print("\nData Sorted by Marks:")
```

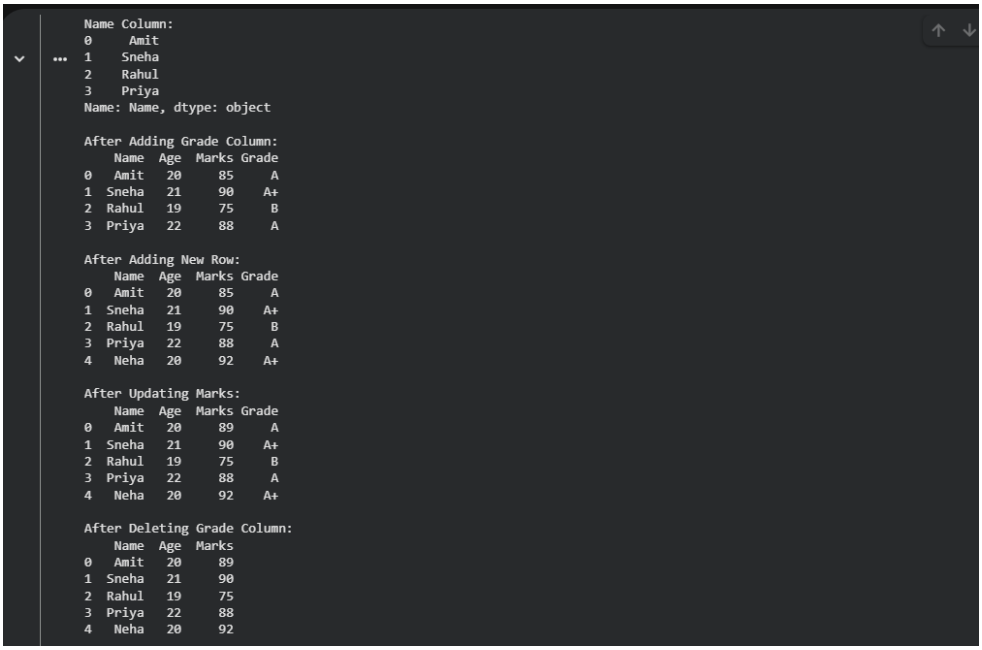
```
print(df.sort_values("Marks", ascending=False))

# 8. Calculate average
print("\nAverage Marks:")
print(df["Marks"].mean())

# 9. Find maximum marks
print("\nMaximum Marks:")
print(df["Marks"].max())

# 10. Find minimum marks
print("\nMinimum Marks:")
print(df["Marks"].min())
```

OUTPUT



```
Name Column:
0     Amit
1    Sneha
2    Rahul
3    Priya
Name: Name, dtype: object

After Adding Grade Column:
    Name  Age  Marks Grade
0   Amit   20     85     A
1  Sneha   21     90    A+
2  Rahul   19     75     B
3  Priya   22     88     A

After Adding New Row:
    Name  Age  Marks Grade
0   Amit   20     85     A
1  Sneha   21     90    A+
2  Rahul   19     75     B
3  Priya   22     88     A
4   Neha   20     92    A+

After Updating Marks:
    Name  Age  Marks Grade
0   Amit   20     89     A
1  Sneha   21     90    A+
2  Rahul   19     75     B
3  Priya   22     88     A
4   Neha   20     92    A+

After Deleting Grade Column:
    Name  Age  Marks
0   Amit   20     89
1  Sneha   21     90
2  Rahul   19     75
3  Priya   22     88
4   Neha   20     92
```

```
Students with Marks greater than 85:
    Name  Age  Marks
0   Amit   20     89
1  Sneha   21     90
3  Priya   22     88
4   Neha   20     92

Data Sorted by Marks:
    Name  Age  Marks
4   Neha   20     92
1  Sneha   21     90
0   Amit   20     89
3  Priya   22     88
2  Rahul   19     75

Average Marks:
86.8

Maximum Marks:
92

Minimum Marks:
75
```